Question 6 **(25 marks)**

(a) **(i)** Differentiate the function $f(x) = 4x^3 - 3x^2 + x - 7$, where $x \in \mathbb{R}$, with respect to x.

(ii) Find the slope of the tangent to the graph of $f(x) = 4x^3 - 3x^2 + x - 7$ at the point $(1, -5)$.

(iii) Hence find the equation of the tangent to the graph at this point.

(b) The function $g(x) = 2x^2 + px + q$, where $p, q \in \mathbb{Z}$, and $x \in \mathbb{R}$.
Given that $g(2) = 6$ and $g'(3) = 9$, find the value of p and the value of q.

Note: $g'(3)$ is the value of the derivative of $g(x)$ at $x = 3$.

$p =$ $q =$

Q6	Model Solution – 25 Marks	Marking Notes
(a) (i)	$f'(x) = 12x^2 - 6x + 1$	**Scale 10C (0, 4, 8, 10)** *Low Partial Credit:* 1 term [other than −7 (see note)] correctly differentiated *High Partial Credit:* 2 terms correctly differentiated **Note**: differentiation of independent term must be explicit for marks to be awarded i.e. $f'(-7) = 0$
(a) (ii)	$12(1)^2 - 6(1) + 1 = 7$ (Slope)	**Scale 5C (0, 3, 4, 5)** *Low Partial Credit:* Any substitution into candidates $f'(x)$ $f'(1)$ or equivalent written *High Partial Credit:* $f'(-1)$ fully substituted
(a) (iii)	$y + 5 = 7(x - 1)$ $7x - y - 12 = 0$	**Scale 5C (0, 3, 4, 5)** *Low Partial Credit:* Eq. of line formula with some substitution *High Partial Credit:* Eq. of line formula fully substituted with $(1, 5)$ but not candidates answer from (a)(ii)
(b)	$g(2) = 2(2)^2 + p(2) + q$ $8 + 2p + q = 6$ $2p + q = -2$ $g'(3) = 4(3) + p = 9$ $\Rightarrow p = -3$ Sub $p = -3$ $\Rightarrow 2(-3) + q = -2$ $q = 4$ $p = -3 \qquad q = 4$	**Scale5D (0, 2, 3, 4, 5)** *Low Partial Credit:* $g'(x)$ found $g(2)$ found *Mid Partial Credit:* $g'(x)$ and $g(2)$ found $2p + q = -2$ *High Partial Credit:* p found

Question 8 **(45 marks)**

A company makes and sells fibre optic cable. It can sell, at most, 200 kilometres of cable in a week. For a certain range of its production the company has found that profit can be modelled using the function:

$$P(x) = 275x - x^2 - 2000, \text{ where } x \leq 200.$$

In the function, x is the number of kilometres of fibre optic cable sold and $P(x)$ is the profit in euro.

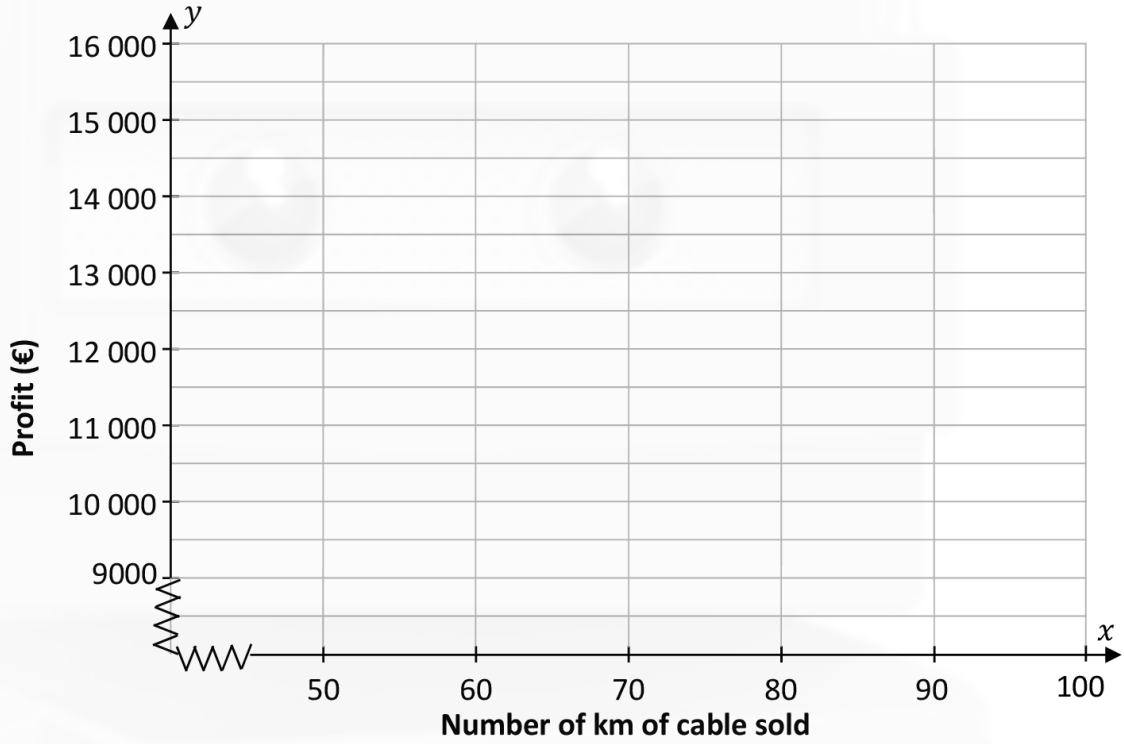
(a) Use the profit function, $P(x)$, to find how much money the company loses if it does not sell any cable.

(b) In a particular week the company made a profit of €8350.
Find the number of kilometres of cable it sold that week.

(c) **(i)** The table below shows some of the data representing the profit on sales.
Use the profit function, $P(x)$, to complete the table.

Number of km of cable sold (x)	50	60	70	80	90	100
Profit (€)			12 350			

(ii) Use the data in the table to draw the graph of the profit function on the axes below for $50 \leq x \leq 100,\ x \in \mathbb{R}$.



(iii) Use your graph to estimate the lower and upper range of sales (in km of cable) in order to make a profit of between €10 000 and €14 000 in a particular week.
Show your work on the graph above.

Lower = Upper =

(d) Use calculus to find the number of kilometres of cable sold when the profit is increasing at a rate of €105 per km.

<table>
<tr>
<td>(a)</td>
<td>€2000</td>
<td>Scale 5B (0, 2, 5)
Partial Credit:
• $x = 0$</td>
</tr>
<tr>
<td>(b)</td>
<td>$275x - x^2 - 2000 = 8350$
$-x^2 + 275x - 10350 = 0$
$x = 45, \ \ x = 230$

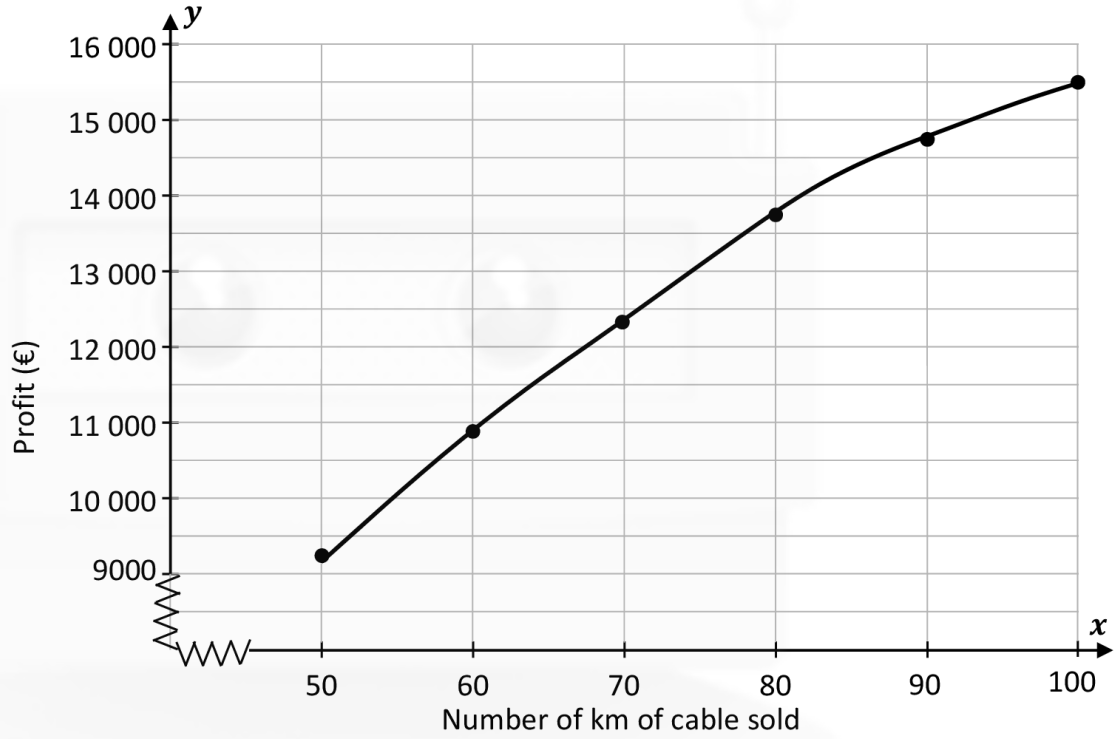
$x = 230$ is not possible as production limit is 200

$x = 45$ km</td>
<td>Scale 10C (0, 4, 6, 10)
Low Partial Credit:
• quadratic equation written
• 45 without work

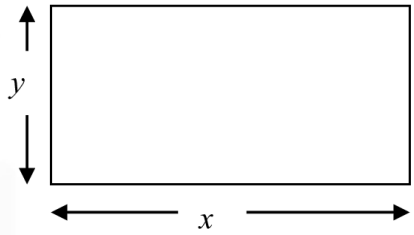
High Partial Credit:
• roots of quadratic

Note: 45 subbed in to get 8350 is full Marks</td>
</tr>
<tr>
<td>(c)
(i)</td>
<td><table>
<tr><td>50</td><td>60</td><td>70</td><td>80</td><td>90</td><td>100</td></tr>
<tr><td>9250</td><td>10 900</td><td>12 350</td><td>13 600</td><td>14 650</td><td>15 500</td></tr>
</table></td>
<td>Scale 10C (0, 4, 6, 10)
Low Partial Credit:
• 1 or 2 correct new entries

High Partial Credit:
• 3 or 4 correct new entries</td>
</tr>
</table>

(c) (ii)		
(c) (ii)	Graph above	**Scale 5C (0, 2, 3, 5)** *Low Partial Credit:* • 1, 2 or 3 correct plots • bar chart *High Partial Credit:* • 4 or 5 correct plots • 6 correct plots but not joined
(c) (iii)	Lower 55 Upper 83 (Lines to be shown on diagram)	**Scale 5B (0, 2, 5)** *Partial Credit:* • evidence of use of graph • one correct range found • correct answers without work
(d)	$\frac{dP}{dx} = 275 - 2x = 105$ $2x = 170$ $x = 85$	**Scale 10C (0, 4, 6, 10)** *Low Partial Credit:* • effort at calculus *High Partial Credit:* • correct differentiation

Kieran has 21 metres of fencing. He wants to enclose a vegetable garden in a rectangular shape as shown.

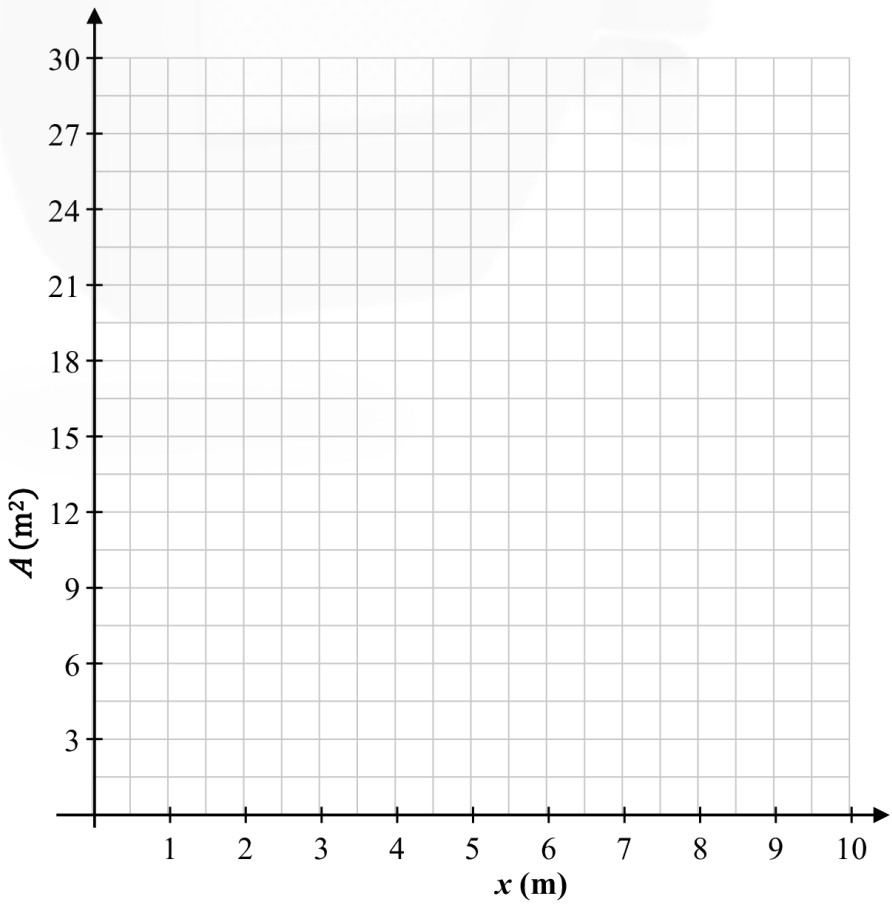


(a) By writing an expression for the perimeter of the vegetable garden in terms of x (length in metres) and y (width in metres), show that $y = 10{\cdot}5 - x$.

(b) **(i)** Complete the table below to show the values of y and A (the area of the garden) for each given value of x.

x (m)	0	1	2	3	4	5	6	7	8	9	10
y (m)					6·5						
A (m^2)					26						

(ii) Use the values of x and A from the table to plot the graph of A on the grid below.



(c) Use your graph to estimate the maximum value of A and write the corresponding length and width.

A: Maximum area (m^2)	
Length (m)	
Width (m)	

(d) **(i)** Show that the area of the rectangle can be written as $A = 10{\cdot}5x - x^2$.

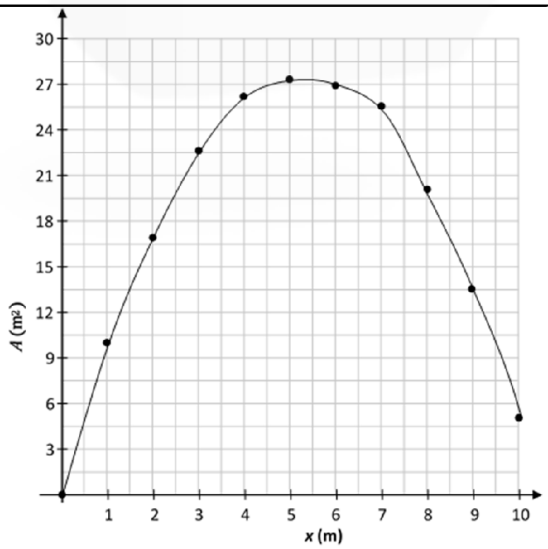
(ii) Find $\dfrac{dA}{dx}$.

(iii) Hence, find the value of x which will give the maximum area.

(iv) Find this maximum area.

<table>
<tr><th>Q8</th><th>Model Solution – 50 Marks</th><th>Marking Notes</th></tr>
<tr><td>(a)</td><td>$$2x + 2y = 21$$
$$x + y = 10{\cdot}5$$
$$y = 10{\cdot}5 - x$$</td><td>Scale 5B (0, 2, 5)
Partial Credit
• Some correct work on perimeter
• Relevant work on diagram</td></tr>
<tr><td>b
(i)</td><td>
<table>
<tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr>
<tr><td>y</td><td>10·5</td><td>9·5</td><td>8·5</td><td>7·5</td><td>6·5</td><td>5·5</td><td>4·5</td><td>3·5</td><td>2·5</td><td>1·5</td><td>0·5</td></tr>
<tr><td>A (m^2)</td><td>0</td><td>9·5</td><td>17</td><td>22·5</td><td>26</td><td>27·5</td><td>27</td><td>24·5</td><td>20</td><td>13·5</td><td>5</td></tr>
</table>
</td><td></td></tr>
<tr><td>b
(i)</td><td></td><td>Scale 5D (0, 2, 3, 4, 5)
Low Partial Credit
• relevant calculations without any correct entry
• 1, 2, 3 or 4 correct entries to table (y or A)

Mid Partial Credit
• 5 or 6 correct pairs of entries (y and A) or full y line correct

High Partial Credit
• 7, 8 or 9 correct pairs of entries (y and A)</td></tr>
<tr><td>(b)
(ii)</td><td> 
</td><td>Scale 5D (0, 2, 3, 4, 5)
Low Partial Credit
• 1, 2, 3 or 4 correct plots
• Graphs y only

Mid Partial Credit
• 5, 6, 7 or 8 correct plots

High Partial Credit
• 9, 10 or 11 correct plots but no joining or incorrect joining</td></tr>
</table>

<table>
<tr><td>(c)</td><td>
<table>
<tr><td>Maximum area</td><td>27·5 ≤ A ≤ 28</td></tr>
<tr><td>x value</td><td>5 ≤ X ≤ 5·5</td></tr>
<tr><td>y value</td><td>5·5 ≥ Y ≥ 5</td></tr>
</table>
</td><td>Scale 5C (0, 2, 4, 5)
Low Partial Credit
• Maximum area identified from graph

High Partial Credit
• Area and one corresponding value</td></tr>
<tr><td>(d)
(i)</td><td>$A = xy$
$A = x(10{\cdot}5 - x)$
$A = 10{\cdot}5x - x^2$</td><td>Scale 15C (0, 3, 7, 15)
Low Partial Credit
• Relevant area formula

High Partial Credit
• Correct substitution into area formula</td></tr>
<tr><td>(d)
(ii)</td><td>$\frac{dA}{dx} = 10{\cdot}5 - 2x$</td><td>Scale 5B(0, 2, 5)
Partial Credit
• Any correct differentiation</td></tr>
<tr><td>(d)
(iii)</td><td>$10{\cdot}5 - 2x = 0$

$x = 5{\cdot}25$</td><td>Scale 5C (0, 2, 4, 5)
Note: Must have calculus in (d)(ii) to get any marks in this section

Low Partial Credit
• Any use of answer to (d)(ii)
High Partial Credit
• Lets their derivative = 0</td></tr>
<tr><td>(d)
(iv)</td><td>$A = xy$
$A = 5{\cdot}25 \times 5{\cdot}25$
$A = 27{\cdot}56$
Or
$A = 10{\cdot}5(5{\cdot}25) - (5{\cdot}25)^2$
$55{\cdot}125 - 27{\cdot}5625$
$A = 27{\cdot}56$</td><td>Scale 5B (0, 2, 5)
Partial Credit
• Any substitution into area formula or equation given in (d)(i)
• Writes area formula</td></tr>
</table>